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Leadership and performance in Japanese R&D teams

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This study examined the relative influence of transformational and gatekeeping leadership on team performance in a study of researchers working in industrial R&D teams in Japan. Potential effects of both internal and external communication and group norms for consensus were studied as possible mediating influences on the leadership-performance relationship. Results found that, while both forms of leadership enhanced communication processes within and between groups, only gatekeeping leadership served to reduce group norms for consensus. As a result, team cultures became somewhat more accepting of expressions of divergent opinions and new ideas from various team members, an important factor in R&D innovation and performance. By contrast, transformational leadership served to create team cultures in which divergence from group norms by various members was discouraged, leading to fewer innovative ideas and no performance increment. Results are discussed both in the context of the unique Japanese work environment and in the larger context of leadership processes across regions and cultures.

Keywords: communication; consensus norms; gatekeeping leadership; Japan; R&D teams; team performance; transformational leadership

Introduction

Technological innovation has long been considered to be one of Japan's principal competitive assets in the highly volatile and high-stakes global economy. Developing new technologies, and new applications for existing technologies, are routinely cited as a principal reason for Japan's lead in markets such as consumer and industrial electronics, chemicals and pharmaceuticals and, increasingly, aerospace (Shibata 2006). At the centre of these efforts are the R&D research laboratories that churn out new products and applications at an unrelenting pace. At the heart of these labs is an approach to knowledge management that blends culture, technology, and process engineering (Nonaka 2005, Osono *et al.* 2008).

As a result of Japan's prominence on the world stage, numerous studies have been conducted over the years in an effort to explicate the principal factors that influence Japan's R&D laboratory performance (Clark and Fujimoto 1991, Nonaka and Takeuchi 1995, Kono and Clegg 2001, Lervik 2008, Iyata-Arens 2009, Motohashi 2009). Most of these studies have focused on the perceived uniqueness of Japan's culture, work ethic, engineering prowess, knowledge management systems, national innovation system, and industrial policies. A small subset of these studies focused specifically on leadership style as a potential influence. Given Japan's relatively strong cultural emphasis on collective behaviour, leadership style represents a potentially critical factor to consider in exploring

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how and why Japanese research teams perform at such high levels. However, even though leadership has long been identified as a key factor in team behaviour and effectiveness in other cultural settings, little is known about its relative impact on innovation and performance in the third largest economy of the world (Shibata 2006).

While the literature on leadership processes is rich in both theory and research, little has focused specifically on research and development teams, preferring other organizational settings such as manufacturing and marketing (Scott and Bruce 1998, Elkins and Keller 2003, Jackson and Tomioka 2004, Aycan 2008). Much of the research that exists tends to focus on the performance implications of *transformational leadership* (Bass and Riggio 2006). The principal finding here is that transformational, or charismatic, leaders can often create conditions in R&D teams that, not unlike other work environments, are conducive to exceptional effort and creativity. This, in turn, can lead to enhanced group performance (Elkins and Keller 2003).

The empirical question to be addressed here, however, is the extent to which transformational leadership (or any approach to leadership) can transcend cultural differences. Specifically, our question is whether transformational leadership, which some evidence suggests can lead to higher performance in Western research laboratories, can work equally effectively in non-Western settings. In point of fact, little evidence exists concerning the relative effectiveness of transformational leadership across cultures, particularly in East Asia (Avolio *et al.* 2005). Research by the GLOBE research project, for example, suggests that such leadership can be effective in some, but certainly not all, cultural environments (House *et al.* 2004, Dickson *et al.* 2009).

Because values and cognitive frameworks are different in collectivistic cultures compared to individualistic cultures, so too might we expect leadership styles to differ (Hooker 2003, Steers *et al.* 2010). For example, Abe (2010) suggested that leadership style, which promoted group oriented decision-making, was effective in Japan. Furthermore, in their systematic study, House *et al.* (2004) and Dickson *et al.* (2009) identified Japan as one environment where the effectiveness of Western-style leadership might be limited. In particular, strong consensus norms might negatively affect individual and organizational performance (particularly in teams) by curbing unique ideas that run counter to prevailing beliefs (Ishikawa 2008). If this is correct, the effects of transformational leadership on such norms should be examined in Japan. This study aims to investigate whether transformational leadership has a positive or negative impact on Japanese R&D team performance through norms for maintaining consensus. Previous studies have neglected this issue largely because it is not a concern in many Western settings. Nevertheless, it may be significant in a collectivistic culture like Japan.

Previous studies of transformational leadership across cultures are also limited in that they have routinely ignored the effects of communication patterns in team performance (Aycan 2008). It is argued here, that communication processes may be more important in R&D settings than other settings due to the creative and sometimes abstract interchanges that are required among team members. In this regard, several studies have found that variations in patterns of communication can significantly affect R&D team performance and knowledge creation (Katz and Tushman 1979, Kivimaki and Lansisalmi 2000, Hirst and Mann 2004, Halevy and Sagiv 2008). However, there have been few studies that examined the relationship between transformational leadership and team communication in R&D settings, and none focusing on Japan. In this study, it is proposed that transformational leadership will have a positive impact on R&D team performance through the team communication process in a Japanese sample. Thus, this study aims to examine the two kinds of influences of transformational leadership on R&D team

performance: a positive influence through a norm for maintaining consensus and a negative influence through team communication.

Finally, this study compares *gatekeeping leadership* with transformational leadership to determine which leader behaviour is more effective in this environment. In contrast to transformational leadership, gatekeeping leadership emphasizes a supervisor's role in building bridges or making connections between employees and their need for information and resources to complete their assignments (Hirst and Mann 2004). Gatekeepers are facilitators, not cheerleaders; supporters, not commanders. Indeed, many effective gatekeepers may have little or no charismatic abilities. Still, they can play an important role in facilitating team effort and performance. In this study, it is proposed that gatekeeping leadership positively influences Japanese R&D team performance through a norm for maintaining consensus and through communication. In other words, it is proposed that gatekeeping leadership is more effective than transformational leadership in influencing independently measured performance in the Japanese context.

Transformational leadership and gatekeeping leadership

Transformational leadership

Transformational leadership has been defined as leader behaviour that influences followers to broaden and elevate their goals and provide them with the confidence to perform beyond the expectations specified in implicit or explicit exchange agreements (Dvir *et al.* 2002, Avolio *et al.* 2005, Dickson *et al.* 2009). Advocates of this model argue that most previous leadership models (generally referred to as transactional models) were premised on assumptions about followers' rational decision-making; that is, followers behaved in order to maximize their rewards in the organization, which required leaders to integrate followers' rewards with their goals. As such, a leader's principal responsibility is to set goals, clarify desired outcomes, provide feedback, and exchange rewards for accomplishments (Dvir *et al.* 2002). While advocates of transformational leadership do not disagree with the value of transactional exchanges, they argue that in reality followers do not necessarily always behave based solely on rational decision-making principles. Particularly in high performance work teams, they assert that followers will often emphasize their contribution to team performance over their rational exchange agreements with the organization.

According to Bass and Avolio (1990), Avolio *et al.* (2005), and Aycan (2008), transformational leadership consists of four factors: idealized influence, inspirational motivation, intellectual stimulation, and individualized consideration. The idealized influence is a behaviour that inspires followers and fosters identification with their leader. The inspirational motivation is a behaviour that encourages followers by expressing enthralling visions. The intellectual stimulation includes drawing followers' attention to problems and promoting novel perspectives and their intellectual curiosity. The individualized consideration involves realizing followers' needs and providing them with support and coaching. These behaviours encourage followers' ardent contribution to the achievement of team goals.

Transformational leadership has been studied extensively in recent years and has been found to be fairly consistently related to positive job attitudes (Podsakoff *et al.* 1990, Dumdum *et al.* 1991, Barling *et al.* 1996) and the job performance of followers (Dumdum *et al.* 1991, Howell and Avolio 1993, Yammarino and Dubinsky 1994, Barling *et al.* 1996, Lowe and Galen Kroeck 1996, Dvir *et al.* 2002). Only a small subset of these studies focused specifically on R&D teams, the principal focus of the present study. For example,

Keller (1992) found that transformational leadership was positively related to R&D team performance, which included project quality and budget/schedule rated by project members and managers, respectively. Transformational leadership significantly influenced project quality and budget/schedule in both research and development teams, although the significance of the correlation was more prominent in research projects than in development projects. Keller (2006) also noted that, based on a longitudinal study, that transformational leadership could predict five-year-later profitability and speed to market.

Berson and Linton (2005) compared the effects of transformational leadership in R&D settings with those in non-R&D settings and found that, although a significant correlation between transformational leadership and outcome variables including quality climate, job satisfaction, and overall satisfaction was observed in both R&D and non-R&D settings, it was stronger in R&D settings than in non-R&D settings. While job attitudes are important for management here, it is also possible that such leadership would also influence R&D performance, although the authors did not specifically study this relationship. Moreover, while Keller (1992, 2006) and Berson and Linton (2005) both found positive relationships between transformational leadership and R&D team performance, they did not search for possible factors that might mediate this relationship.

Finally, Shin and Zhou (2003) found that intrinsic motivation mediated the relationship between transformational leadership and the creativity of followers in Korean R&D settings, and that followers' collectivistic norms moderated the relationship between them. They indicated that these norms largely affected the efficiency of transformational leadership in the Korean work environment. However, since Shin and Zhou (2003) focused their study on an individual (as opposed to a group) level, it is difficult to generalize to groups. Moreover, some caution is in order here (and throughout this study) about over-generalizing about which style of leadership is most effective across cultures as significant differences can sometimes be found (Paik and Pak 2009).

The results of these studies indicate that transformational leadership was positively related to R&D team performance in the samples studied. Still, the question remains concerning both the external generalizability of these results (specifically, the potential applicability of these results to the Japanese work environment) and possible mediators of this relationship. Two mediators, a norm for maintaining consensus and communication patterns, are examined in this study.

Norm for group consensus

As mentioned above, some previous studies showed transformational leadership can positively influence R&D team performance in Western settings. Nevertheless it is possible that transformational leadership also has a negative impact on R&D team performance through the strong norm for maintaining consensus in Japanese settings. Japan is a collectivistic and high context culture in which group consensus is highly valued (Dulek and Fielden 1991, Kono and Clegg 2001, Abe 2010). Local management practices are designed to promote this, even in R&D divisions (Hara *et al.* 2008). Unlike in Britain, Germany, or the US, most Japanese R&D employees are recruited as new college graduates, and tend to devote themselves to a single company until retirement. Moreover, company-specific knowledge and skills are emphasized, and in-house training is regarded as important (McCormick 1995). Although Japanese companies nowadays are changing some of their traditional HRM practices to Western style practices, these shifts are occurring very slowly and cautiously. For example, performance-based compensation systems have been applied to limited hierarchies in the area of R&D (Benson and Debroux

2003, Hemmert 2009). Because of this, intellectual inbreeding often occurs and radical innovations may fail to emerge at a rate often found in the West.

It is possible that transformational leadership serves to reinforce these norms of group consensus that reinforce either the status quo or only marginal or evolutionary change or innovation. One reason is that transformational leadership has a strong impact on followers. Sometimes the impact is too strong for followers to confront their leaders' opinions. In particular, idealized influence, which is one dimension of transformational leadership, may have such an effect. This idealized influence promotes followers' identification with their leader. The result is very similar to that of charismatic leadership because the primary influence process of charismatic leadership is personal identification, which is the influence derived from followers' desire to please and imitate leaders (Conger 1989). Under such an influence, followers do not want to criticize their leaders and do not permit colleagues to criticize their leaders. From a review of previous studies, Yukl (2002) noted that being in awe of the leader can reduce good suggestions by followers and that a desire for leader acceptance can inhibit criticism by followers and be considered a possible dark side to charismatic leadership.

Other studies indicated that transformational leadership promotes cohesiveness of the team (Jung and Sosik 2002, Pillai and Williams 2004, Stanko and Gibson 2009). This is because transformational leadership fosters followers' commitment to the goal and influences followers to strive for the achievement of the goal in unison with other team members. In general, cohesiveness should have a positive impact on team performance, but excessive cohesiveness may also negatively impact performance by repressing opposing views or ideas. For example, Leana (1985) found that group cohesiveness was a major cause of groupthink. Therefore, transformational leadership may suppress some ideas and opinions expressed by team members.

In addition, transformational leadership often unites followers and creates a team climate, which is required for achievement of the goal (Liao and Chuang 2007). As a result, it may press followers not to behave contrary to the climate the leader aims to create. As a result, team members may be reluctant to diverge from prevailing norms; and indeed, they may even be punished for this. While similar norms are found in the West, the prevailing strength of consensus norms in Japan make it much more difficult for employees to take issue with prevailing ideas, oftentimes preventing them from opposing any consensus reached by the group or organization they belong to. As a result, many new ideas that do not quickly receive group approval can languish or die. For example, Postmes *et al.* (2001) found that strong norms for maintaining consensus adversely affected the quality of group decision making, and led to poor team performance. At the same time, such norms often reduce the diversity of information and opinions within a team, which is essential for team performance (Pelz and Andrews 1966, Halevy and Sagiv 2008).

Therefore, it can be posited that a strong norm for maintaining consensus, which undermines a diversity of new ideas, can eventually hinder team performance in the Japanese context.

Hypothesis 1: A norm for maintaining consensus mediates the relationship between transformational leadership and team performance in such a way that transformational leadership is positively related to a norm for maintaining consensus and this norm for consensus is negatively related to team performance.

Internal and external communication

For R&D teams to work together effectively, open, timely, and frank communication is essential (Menzel 1966, Halevy and Sagiv 2008, Stanko and Gibson 2009). While team members acquire technological information through books and papers, they also make extensive use of external communication with specialists outside of their organizations. Therefore, following the findings of Ancona and Caldwell (1992), we would expect external communication to affect team performance through promoting the acquisition of technological information.

However, information required for R&D performance is not only technological in nature. For example, it is also necessary for team members to acquire information about the manufacturing process and customer needs. In order to acquire this information, internal communication, or communication among members inside the organizations (e.g. members of manufacturing or marketing departments) is also useful. In particular, some studies indicate that high context communication inside organizations fosters knowledge creation in Japanese companies (Nonaka and Takeuchi 1995, Hentschel and Haghirian 2010). Therefore, following the consistent previous findings of the research, we would expect that ample availability of both external and internal communication to be positively related to R&D team performance (Kivimäki and Lansisalmi 2000, Hirst and Mann 2004).

In theory, at least, transformational leadership is, or ought to be, related to team communication. Transformational leadership clarifies the goal of followers (Nemanich and Keller 2007) and promotes goal commitment from them (Piccolo and Colquitt 2006). Such goals make followers clear about what information is necessary or important. New knowledge, which is the goal of R&D activities, is thus created by the composition and interpretation of available information. Furthermore, high commitment to these goals stimulates followers' information seeking activities because information is one of the most important resources for the achievement of team goals.

Individualized consideration, which is one of the four dimensions of transformational leadership, provides followers with support and coaching. Because of this, transformational leadership should have a positive impact on the development of followers (Dvir *et al.* 2002). This includes the advancement of communication skills and inducements for increased communication, since communication is a key factor in R&D performance. Under such influence from transformational leadership, the followers should – at least in theory – improve their communication patterns.

Intellectual stimulation, which is also one dimension of transformational leadership, encourages followers to promote novel perspectives and open their intellectual curiosity. Therefore, they should apply new ideas or try new methods in order to promote novel perspectives and stimulate their intellectual curiosity. Eventually, they should come to communicate more effectively because communication is an important tool for obtaining new ideas or methods.

Transformational leadership has also been found to positively influence leader-member exchange (Hui *et al.* 2005). Followers, who have good relationship with their leaders, desire to meet their demands. In this case, the demands from leaders typically involve goal achievement and the acquisition of information. Further, the ideal influence dimension contributes to followers' having more than just self-interest in mind, thus overcoming the fear of incurring personal costs in conducting various communication activities.

Hence, followers under transformational leadership should search for information that is required to achieve their goals. In this regard, Madzar (2001) found that

transformational leadership was positively related to the information inquiry behaviours of followers, at least with respect to followers' behaviours that sought information from their leaders. However, as noted above, other types of information are also required to achieve team goals, including outside information. Accordingly, we posit that transformational leadership helps to promote both internal and external communications of followers:

Hypothesis 2a: Internal communication mediates the relationship between transformational leadership and team performance in such a way that transformational leadership is positively related to internal communication and internal communication is positively related to team performance.

Hypothesis 2b: External communication mediates the relationship between transformational leadership and team performance in such a way that transformational leadership is positively related to external communication and external communication is positively related to team performance.

Gatekeeping leadership

Gatekeeping leadership is a construct that has evolved from the more general construct of a gatekeeper. A number of empirical studies have focused on the role of gatekeepers in R&D team performance (Taylor and Utterback 1975, Hirst and Mann 2004). In particular, gatekeepers tend to be 'communication stars' that can perform both internal communication and external communication roles frequently and effectively. This is not necessarily easy to accomplish, since both contexts are different from each other. Nevertheless, gatekeepers can often perform both communication activities simultaneously and effectively. Indeed, they often perform important roles as a node in highly diverse information flows. Gatekeepers often have communication networks with specialists outside the organizations. In this sense, they bridge organizational boundaries. They not only acquire the technological information from the networks, which is required for R&D teams, but also translate such information into terms that are meaningful to other team members within an internal context. They also communicate with employees of other departments inside the organizations. Previous studies showed that in R&D teams gatekeepers positively influenced performance by encouraging communication with other team members.

Meanwhile, Allen (1977) suggested that gatekeepers were not always equivalent to team leaders. That is, leaders and gatekeepers may exist separately and play different roles in teams. However, if leaders and gatekeepers exist separately, the flow of directions and information may get confused. On the contrary, when leaders play the role of gatekeepers simultaneously, smoother communication within the team would be performed and team performance may increase.

In the present study, gatekeeping leadership is defined as leadership that plays the role of gatekeeper. In the teams where leaders display gatekeeping leadership, team members may actively perform communication and eventually have high performance. In this regard, Hirst and Mann (2004) show that leaders that performed the role of gatekeepers positively influenced R&D team performance. Furthermore, Kim and Min (1999) found that the gatekeeping role was one of the most important roles for R&D team leaders in a Korean sample. This suggests that gatekeeping leadership may be effective in collectivistic cultures.

Internal and external communication

Several early studies show that gatekeepers could influence the communication patterns and richness of other members in R&D teams (Allen 1977, Tushman 1977, Katz and Tushman 1979, Tushman and Katz 1980). For example, Tushman and Katz (1980) showed that gatekeepers influenced team performance through encouraging the external communication of other team members, while Tushman (1977) found that the gatekeeping role affected the internal communication of other members. Further, Harada (2003) noted that gatekeepers influenced the communication of other members in Japanese R&D settings. Finally, Tushman and Katz (1980) suggested that non-supervisor gatekeepers have a negative association between external communication and performance, although supervisor gatekeepers have a positive association. The results of these studies suggest that gatekeeping leadership positively influences the internal and external communication of team members.

Furthermore, gatekeeping leaders actively perform both external and internal communication because they understand that communication is critical for R&D performance. As a result, leader expectations have a significant impact on the learning behaviour of followers through goal setting, providing learning opportunities, and feedback (Bezuijen *et al.* 2009). Accordingly, if gatekeeping leaders offer goal setting for communication, communication opportunities, and feedback for team members' communication, team members come to perform communication more actively.

Because of these reasons, team members with gatekeeping leaders communicate more positively with internal or external individuals. Frequent communication promotes information sharing, such as technological information and internal information that includes the vision and strategy of the company, information about the manufacturing process, customer needs, and other information. Based on this, the following hypotheses are proposed:

Hypothesis 3a: Internal communication mediates the relationship between gatekeeping leadership and team performance in such a way that gatekeeping leadership is positively related to internal communication and internal communication is positively related to team performance.

Hypothesis 3b: External communication mediates the relationship between gatekeeping leadership and team performance in such a way that gatekeeping leadership is positively related to external communication and external communication is positively related team performance.

Norm for group consensus

Gatekeeping leaders bring various types of information to teams. Because of this, team members can share various types of information inside the teams. This means that they share various opinions, various values, and various points of view. Such sharing reduces the pressure that builds up consensus norm. Gatekeeping leaders understand that using various contexts is critical for R&D performance. Hence, they attach importance to the diversity of opinions, values, and points of views because such diversity is important in various contexts. Because of this, gatekeeping leaders would try to shape norms that regard diversity as important. Taggar and Ellis (2007) showed that leader expectations had a significant impact on norm creation. In this regard, Feldman (1984) found that norms are enforced when they facilitate team survival. Conversely, norms are weakened if team

members recognize that they do not contribute to team survival. Team members thus weaken norms for maintaining lock-step consensus because they regard diverse opinions to be important for team survival.

Moreover, it is possible that gatekeeping leadership facilitates communication among team members. It fosters team member behaviour that actively exchanges opinions inside the teams. As a result, team members share various opinions. The members in such teams do not shape consensus norms because they share and respect various kinds of opinions.

Because of these reasons, team members under gatekeeping leadership reduce norms for maintaining consensus. As mentioned above, norms for maintaining consensus would impede team performance. Hence, it is posited that:

Hypothesis 4: A norm for maintaining group consensus mediates the relationship between gate-keeping leadership and team performance in such a way that gate-keeping leadership is negatively related to norm for maintaining consensus and a norm for consensus is negatively related to team performance.

Method

Sample

The data for this study were obtained through a survey carried out during October 2007 and March 2008. The sample consisted of 122 R&D teams from seven industrial parts manufacturers in Japan. In addition to 122 team leaders (response rate: 100%), 683 R&D team members (response rate: 82.4%) and 25 managers (response rate: 100%) also participated in the study. Each team member belonged to a single team, and each team was managed by one of the 25 managers. Among the team leaders, 94.1% were men, 9.3% held a doctoral degree, and their average age was 36 years old. Among the team members, 89.1% were men, their average age was 30 years old, and their average tenure in the team was two years. The average team size was six persons.

Distribution and collection of the questionnaires was done on company premises and with the prior approval of the R&D department management of each firm. Because of this, response rate was high. The questionnaires were collected by the author in sealed envelopes, therefore confidentiality was maintained.

Measures

This study consisted of six questionnaire measures, plus control variables. Corresponding Japanese versions for all of the measures used in this study were constructed in accordance with the translation-back-translation procedure outlined by Brislin (1980).

Transformational leadership was measured by 20 items adapted from Bass's Multifactor Leadership Questionnaire (MLQ) Form 5X (Bass and Avolio 2004). Each team member was asked to rate his/her leader's behaviour on a five-point response scale. Four items were used to measure inspirational motivation, intellectual stimulation, and individualized consideration. Eight items were used to measure idealized influence. In order to verify that these four factors did contribute to an overall transformational leadership index, a confirmatory factor analysis was conducted. The result of the analysis suggested that a higher-order factor solution provided an adequate fit ($\chi^2 = 131.27$ ($p < 0.05$), AGFI = 0.91, CFI = 0.94, RMSEA = 0.04, NFI = 0.93).

Gatekeeping leadership was measured by five items adapted from Hirst and Mann (2004). An example of the items used in this scale is 'The leader coordinates the team's task efforts with outside stakeholders'. Each team member was asked to rate his/her leader's behaviour on a five-point scale, ranging from strongly agree to strongly disagree.

Internal and external communication were measured by three items each. Each team member was asked to rate the frequency of internal and external communication in his/her team on a five-point response scale, ranging from strongly agree to strongly disagree.

The norm for maintaining consensus was measured by three items adapted from Postmes *et al.* (2001). Each team member was asked to rate his/her group norm for maintaining consensus on a five-point response scale, ranging from very strong to very weak. For example, one of the three measuring items was 'People in this group generally adjust to one another with ease'.

A confirmatory factor analysis was conducted in order to see if transformational leadership, gate-keeping leadership, internal communication, external communication, and norm for maintaining consensus were the distinct construct in each category, respectively. The result of the analysis suggested that a five-factor solution had a better fit than other solutions.

Team performance was measured independently by four items adapted from Keller (2001); that is, technical quality, schedule performance, cost performance, and overall team performance. The managers, who are higher in rank than team leaders, were asked to rate each item on a five-point response scale.

Control variables include team size, which was provided by the leaders, and team members' average tenure provided by the team members.

Aggregation tests

Transformational leadership, gatekeeping leadership, internal communication, external communication, and norm for maintaining consensus were aggregated to mean values within each team, which was the unit of the analysis. To justify this aggregation, a within-group correlation (r_{wg}) was computed to assess the amount of agreement by the team members (James *et al.* 1984). The mean r_{wg} value was 0.87 for transformational leadership, 0.89 for gatekeeping leadership, 0.90 for internal communication, 0.89 for external communication, and 0.89 for norm for maintaining consensus. In addition, the ICC(1) values were as follows: transformational leadership, 0.22; gatekeeping leadership, 0.50; internal communication, 0.24; external communication, 0.36; and norm for maintaining consensus, 0.25. ICC(2) values were as follows: transformational leadership, 0.61; gatekeeping leadership, 0.85; internal communication, 0.64; external communication, 0.76; and norm for maintaining consensus, 0.65. The overall pattern of results across the r_{wg} , ICC(1), and ICC(2) analyses provided sufficient support for aggregating the data to team level of analysis.

Results

The mean values, standard deviations, and coefficient alphas, as well as a correlation matrix are shown in Table 1. As expected, transformational leadership had significant positive correlations with internal communication and a norm for maintaining consensus for this Japanese sample. However, it was not related to external communication or, perhaps most importantly, team performance. Contrarily, gatekeeping leadership showed positive correlations with internal communication, external communication and, again

Table 1. Descriptive statistics and correlations.

	Mean	s.d.	α	1	2	3	4	5	6	7
1. Team size	5.59	1.21								
2. Team tenure: Member	1.94	1.14		0.02						
3. Transformational leadership	3.15	0.59	0.93	0.01	0.04					
4. Gatekeeping leadership	3.36	0.75	0.81	-0.10	-0.02	0.23*				
5. Norm for maintaining consensus	2.38	0.72	0.81	0.16	0.19*	0.37**	-0.20*			
6. Internal communication	3.80	0.53	0.80	-0.04	0.18*	0.43**	0.54**	0.14		
7. External communication	2.50	0.84	0.85	-0.09	-0.21*	-0.04	0.39**	-0.41**	0.09	
8. Team performance	3.01	0.91	0.84	-0.12	0.12	0.17	0.32**	-0.24**	0.37**	0.17

Note: * $p < 0.05$, ** $p < 0.01$.

importantly, team performance. Gatekeeping leadership was also negatively correlated with a norm for maintaining consensus.

To test the relationship between leadership and various potential mediators, structural equation modelling was conducted; the results are shown in Figure 1. As expected, this analysis found a significant positive correlation between transformational leadership consensus norms. Furthermore, transformational leadership was positively related to internal communication, but not to external communication. In contrast, gatekeeping leadership was negatively related to a norm for maintaining consensus and positively related to internal and external communication. In addition, there was a negative correlation between the norm for maintaining consensus and team performance and a positive correlation between internal communication and team performance. However, there was no significant correlation between external communication and team performance. These results suggest that external communication does not mediate between the two kinds of leadership and team performance. Moreover, this model did not show adequate fit ($\chi^2 = 38.09$ ($p < 0.05$), AGFI = 0.82, CFI = 0.87, RMSEA = 0.11, NFI = 0.80).

Hence, another model that excluded external communication was examined. Results are shown in Figure 2. This model shows an adequate fit ($\chi^2 = 20.25$ ($p > 0.05$),

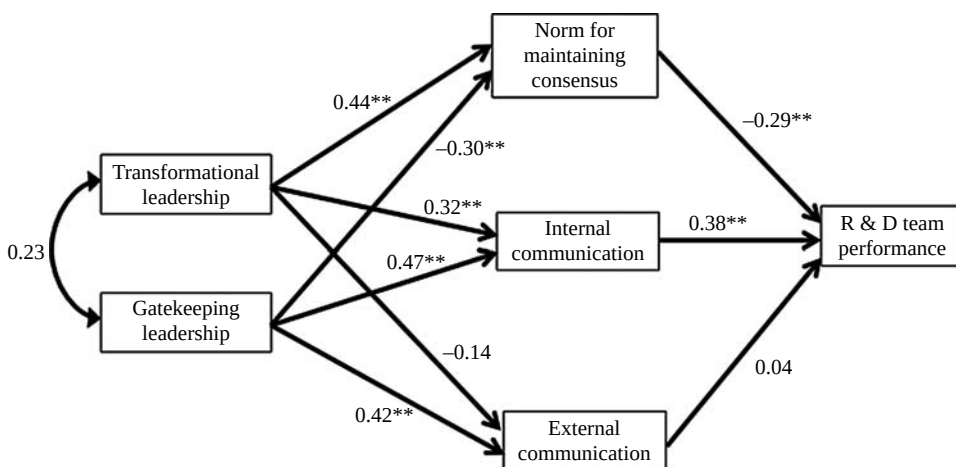


Figure 1. Result of SEM^a.

Note: ** $p < 0.01$, ^aThe effect of control variables are not shown.

AGFI = 0.89, CFI = 0.94, RMSEA = 0.08, NFI = 0.87), and all the paths shown in it were significant. A negative correlation was found only between a norm for maintaining consensus and team performance, while the other correlations were all positive. The result of this analysis suggests that the norm for maintaining consensus mediates the relationship between transformational leadership and team performance in such a way that transformational leadership was positively related to the norm for maintaining consensus, while the norm for maintaining consensus was negatively related to team performance. The result also indicates that the norm for maintaining consensus mediates the relationship between gatekeeping leadership and team performance in such a way that gatekeeping leadership was negatively related to the norm for maintaining consensus and the norm for maintaining consensus was negatively related to team performance. Meanwhile internal communication positively mediated the relationship between both leadership and team performance.

Although Figure 2 suggests an indirect relationship between leadership and team performance, it is possible that there is a direct relationship between them. Therefore, a partially mediated model, which was obtained by adding a direct path between both leadership and team performance, was conducted. The result of this analysis indicated an adequate fit ($\chi^2 = 17.16$ ($p > 0.05$), AGFI = 0.88, CFI = 0.95, RMSEA = 0.08, NFI = 0.89), but correlations between both leadership and team performance were not significant.

These results support Hypotheses 1, 2a, 3a, and 4, but not Hypotheses 2b and 3b.

Discussion

In this study, two different approaches to leadership (transformational and gatekeeping) were compared as they relate to independently measured job performance in Japanese R&D teams. The results indicate that gatekeeping leadership was directly related to team performance for this sample, while transformational leadership was not. As hypothesized, gatekeeping leadership enhanced internal communication among team members, while simultaneously tempering a norm for group consensus that could limit open discussions and disagreements – a typical success factor in research and development facilities. By contrast, transformational leadership served to increase both communication norms and norms for group consensus, and this combination led to less than optimal performance results. That is, a strong norm for maintaining group consensus restrained the critical analysis of various team members and reduced the overall quality of decision making as a team. Accordingly, transformational leadership affected team performance positively through internal communication, but negatively through the maintenance of a norm for

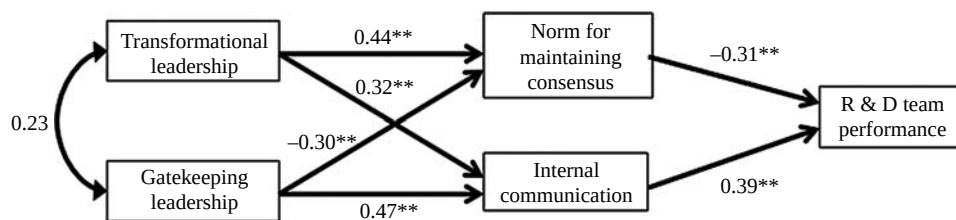


Figure 2. Result of SEM^b excluded external communication.

Note: ** $p < 0.07$, ^bThe effect of control variables are not shown.

maintaining consensus, thereby yielding poor performance results. These findings are consistent with previous observations about knowledge management techniques in Japanese firms (Nonaka 2005, Osono *et al.* 2008).

According to theory, gatekeeping leaders encourage their followers to communicate frequently, even to the point of strong disagreements when it is in the interest of attaining project results (Hirst and Mann 2004). They understand that diverse information and opinions are necessary for research performance. Hence, it is likely that gatekeeping leaders understand not only various types of information processing, but also various cognitive frameworks and personal values that are necessary for R&D team performance and knowledge creation. Under such leadership, team members recognize that various contexts are required for team performance, and they feel it is acceptable to offer their own opinions and suggestions, even if these include criticism of the ideals of colleagues.

With these findings in mind, three study limitations must be identified. First, due to access limitations, this study was not designed as a comparative empirical study. Instead, the data collected here from Japanese employees were compared against reasonably consistent findings of similar employees from various Western studies (Avolio *et al.* 2005). Additional investigations would be helpful that incorporated samples from two or more cultures. This issue can be raised for studies of effective leader behaviour in general. That is, much of the existing research on leadership assumes that these theories are largely culture-free. The present findings call this generalization into question, inviting further studies along these lines.

A second potential limitation is that the present study focused on R&D teams. While a better understanding of leadership processes in such teams is clearly useful, it is not certain that employees and teams in other Japanese organizations or professions would respond in a similar manner. In view of the widespread collective nature of Japanese society, it is not unlikely that many other people in many different work environments would indeed respond in a similar way (Kono and Clegg 2001). This is an empirical question that the present study does not answer.

A third potential limitation of the present study is that the interrelated functions of research and development were not examined separately. A prior study by Keller (1992) suggested that the difference between research and development could be an important moderator in terms of the correlation between transformational leadership and R&D team performance. However, in the present study, companies did not differentiate between these two groups; indeed, the two groups worked as integrated team members, making differentiations impossible without losing the very concept of the teams that were under study.

Implications

It is believed that this study makes some contributions to the theory and practice. First, it raises questions concerning the universal applicability of transformational (or any other) leadership style. Most previous studies concluded, or simply argued without evidence, that transformational leadership had a positive effect on performance regardless of the situation or environment due largely to the uniform influence of charisma on subordinates (Bass and Roggio 2006). The results of the present study do not support this position. Moreover, previous studies have neglected to investigate why transformational leadership may not work equally across cultures. The present study found that transformational efforts adversely affected team performance through their reinforcement of norms for

maintaining group consensus, at least in this Japanese setting. That is, transformational processes led employees to pull together in ways that inhibited the free or independent expression of ideas that may have been at odds with the group. This, in turn, led to fewer ideas or innovations being suggested, which resulted in poorer performance. Although this sequence of events may not have occurred in many Western work environments, its existence in a Japanese work environment is not surprising (Kono and Clegg 2001, Steers *et al.* 2010). This finding suggests that transformational leadership may have a negative impact on team performance in not only Japanese settings but also other collectivistic country settings.

Second, this study examined communication as the mediator between transformational leadership and team performance. Many prior studies indicated that communication had significant impact on R&D performance, and some studies showed that transformational leadership was effective in R&D settings. Nonetheless, there have been no previous studies that examined the relationship between transformational leadership, communication, and R&D team performance. The present study examined this relationship. Although transformational leadership had a negative impact on R&D team performance, it also had a positive impact on internal and external communication.

Third, the present study suggests that gatekeeping leadership is more effective than transformational leadership in collectivistic cultures such as that in Japan. In this sample, transformational leadership had a negative effect on R&D team performance, although it had a positive effect on consensus norms. In contrast, gatekeeping leadership influenced R&D team performance through a combination of a positive influence on group communication and a negative, or tempering, influence on rigid consensus norms. When leaders openly encouraged the frank expression and exchange of ideas that may have run counter to group norms, employees presumably felt more comfortable engaging in such behaviours, resulting in new ideas and subsequently improved performance.

The participants of this study were exclusively R&D employees. Hence, any generalizations of the effects of gatekeeping leadership are limited. However, communication may be a key factor for teams that need creativity beyond just R&D teams. Furthermore, norms for maintaining consensus may represent a major impediment to creativity in teams in general. Because gatekeeping leadership promoted the communication of team members and tempered the consensus norm in this study, it is likely that gatekeeping leadership has a positive impact on other types of teams in other collectivistic countries, compared to transformational leadership.

Conclusions

This study offers evidence concerning the lack of a one-size-fits-all theory of leadership around the globe. Instead, it suggests that future leadership research – as well as future managerial efforts to lead subordinates – must be tempered with a clear recognition of cultural and other local factors that can influence subordinate attitudes and behaviours. This study also suggests that leadership effectiveness in Japan, and possibly other Asian cultures, represents a fruitful area for future research. In point of fact, with few notable exceptions (e.g. Nonaka 2005, Osono *et al.* 2008), little theory-building research focusing on leadership in Asia can be found, despite the growing economic and political importance of this region. Because of this, rigorous research of a comparative nature, both within (e.g. Japan vs. Korea vs. China) and between Asia and other regions (e.g. Asia vs. North America vs. Europe), would prove useful for expanding our collective global understanding of influence processes around the world. In this endeavour, it is hoped that this study represents one useful step.

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